

Three user tests of the Graphene Transport Explorer (**transim**) a developer’s assessment, a student, and a researcher — and what was fixed after each

DSW plug-in **transim** — test report

9 October 2026

Abstract

The Graphene Transport Explorer combines a field model (local Ohmic FEM or a viscous electron fluid) with a semiclassical trajectory Monte Carlo for graphene devices. It was tested three times, each test followed by fixes. In the first test the developer looked for anything that would confuse or mislead a user. In the second, a synthetic first-year PhD student, “Sofia Vanessa” (Politecnico di Milano), explored the program to learn transport. In the third, a synthetic senior experimentalist, “Prof. Jonnson”, checked whether its numbers could go into a publication. Both personas were played by independent agents. They drove the live program in a browser or compiled their own harnesses against its core. They did not edit the program, and every finding comes with a reproduction.

The three tests produced 12, 24 and 15 findings. The most consequential were in the trajectory statistics:

- two random seeds shared 97 % of their trajectories, so seed-to-seed error bars came out about half size;
- a fixed 240 μm path budget cut the diffusive transmission by half at $\ell = 20\text{ nm}$, without warning;
- multi-terminal resistances had no error bars at all;
- the FEM declared itself converged at 21 times its tolerance, giving a conductance four times too large on a long bar.

All four are fixed, and each fix has a gate that fails on the old behaviour. The field model already passed every textbook check (Sharvin to four digits, van der Pauw to 0.3 %, grid-converged R_{xx} and Hall resistance). After the fixes the trajectory model reproduces Landauer–Drude from $\ell = 20\text{ nm}$ to $5\text{ }\mu\text{m}$ within 6 %, and its reported error bars agree with the actual seed-to-seed scatter.

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1 How the tests were run

- **Program under test.** The DSW plug-in `transim` (native C++ core, browser panel) on a local DSW host, with the twelve example devices built the day before: Hall bar, two-terminal strip, van der Pauw square, three EMR geometries, hydrodynamic vicinity and aperture, Veselago lens, two Dirac-fermion-microscope set-ups and a Klein-tunnelling magnetometer.
- **Persona tests.** Each persona was an independent agent with a written brief: who the person is, what they want from the program, and what to look for. Sofia drove the panel in headless Chrome over the DevTools protocol and looked at screenshots. Prof. Jonnson also compiled C++ harnesses against the core to measure numbers against textbook limits. Neither edited the program. Both wrote a report with a narrative in the persona’s voice and numbered findings, each with steps, evidence and a suggested fix. They were asked to re-check every finding and to separate harness mistakes from program faults; both reported harness mistakes of their own.
- **Fixes and evidence.** Each finding was fixed, documented, or deliberately left open (see the status columns below). Every fix to behaviour got a check that fails on the old code. The gates after the last fix:
 - C++: `scatgate` (including bit-identity with the old engine under the legacy hooks), `probe`, `hydro`, `hydrocheck`, `hallcheck`, `veselago`, `dfm` and the new `statcheck`;
 - live panel probes: `layout`, `chart`, `sweepbar`, `map`, `display`, `contacts`, `hydro`, `panel2`, `veselago`, `examples` and the new `tests`.

All are clear.

2 Test 1: the developer’s own assessment

The question was: what in the program would confuse or mislead a user? Table 1 lists what was found and changed.

Table 1: Test 1 findings.

Finding	What was done	Status
FEM Hall voltage read half a cell inside the edge (0.86–0.95 of B/ne)	An edge probe now reads the potential at the edge by extrapolation. A long bar gives $R_{xy} = B/ne$ to 1.0000; the default $4 \times 2 \mu\text{m}$ bar gives 0.959, which is real end-contact shorting at $L/W = 2$.	fixed
My own earlier claim that the mid-column current reads 1.2% low	Wrong: that was measured with the probes still inside the edge. With edge reading, the mid-column flux is exact, and it is the default again.	retracted
Monte Carlo transmissions had no error bars	Every transmission now carries the counts and the binomial $\sigma = \sqrt{T(1-T)/N}$. The table shows $T \pm \sigma$ and flags entries resting on fewer than 100 carriers.	fixed
A loaded density map silently overrides the gate	The sweep strip says so.	fixed

Finding	What was done	Status
With “ ℓ from mobility” off, the FEM uses μ but the trajectories use ℓ	Warning under the field; the hydrodynamic examples derive ℓ from μ .	fixed
BiCGSTAB diverged in the EMR regime	Direct sparse-LU fallback.	fixed
A missing chart value was drawn as zero (<code>isFinite(null)</code> is true)	Uses <code>Number.isFinite</code> .	fixed
Duplicate run and export buttons, three names for two models	One run row; “Field model (FEM / Hydro)” and “Trajectories” throughout.	fixed
Temperature hidden under “Scattering”	Moved beside B .	fixed
Transmission/Validity output went to a clipped panel	Detail panel shown on those two tabs only.	fixed
The 2D map was a bare grid	Map panel with any two of V_g , n , B , T , ℓ , μ , colour bar, hover, CSV.	fixed
Off-centre EMR text claimed an asymmetry a mirror-symmetric bar cannot have	Text corrected; a genuinely off-centre disc example added.	fixed

3 Test 2: Sofia Vanessa, first-year PhD student

3.1 Her account (excerpts)

“The examples were the best part. [...] In two of them the mean free path is given as ‘about $1.7\mu\text{m}$ ’ while the program itself says $0.58\mu\text{m}$ for the same numbers; I spent ten minutes on my calculator before I trusted the program over the text. [...]

Then I got confused, several times, by things that were left over from what I did before. After loading the hydrodynamic example the chart still showed the Hall bar’s Hall resistance [...]. When I loaded an example while a sweep was running, the old sweep simply froze at ‘22/41’ next to the new example’s description. I would have written that into my notes as a result. [...]

On the plain Hall bar, with no viscosity at all, Vorticity showed a dramatic red-and-blue turbulence over the whole device. Only by opening ‘Display & colours’ did I see the colour range was $\pm 8 \times 10^{-9}$ — i.e. nothing. I would have believed it. [...]

The van der Pauw square was where I lost trust: the trajectory curve for R_{xx} goes from +256 to -61Ω and back [...] with no error bars.”

Figure 1 shows the vorticity case: noise at the 10^{-16} level, relative to the current density per cell, stretched to full colour scale.

3.2 Findings and what was done

Table 2: Test 2 findings (severity in brackets).

#	Finding	Fix / evidence	Status
1	(bug) Single runs listed source→probe as 0/2000, read as “nothing reaches the Hall probe”	Probes are transparent in a single run; their rows (zero by construction) are no longer listed.	fixed

#	Finding	Fix / evidence	Status
2	(bug) Results not cleared or marked stale when the example or the settings change	Loading a project stops and clears; any setting change marks the results stale until the next run.	fixed
3	(bug) Loading an example during a sweep left it frozen at 22/41	Load sends Stop and clears the sweep (live probe).	fixed
4	(bug) FEM reported converged at residual 2.1×10^{-6} (tolerance 10^{-7}), G four times too large on a 4000 μm bar	“Converged” now means residual $\leq$ tolerance, otherwise the direct solver takes over. The 4000 μm bar now gives $\sigma W/L$ to 0.8 % (statcheck).	fixed
5	(misleading) Kinks in EMR $R(B)$ at tolerance 10^{-7}	Default tolerance 10^{-9} , with the direct fallback. EMR examples are even in B (examples probe).	fixed
6	(misleading) No error bars on sweeps; vdP R_{xx} down to -61Ω , KT asymmetry $2.5\times$ from noise	Jackknife error over 10 trajectory batches on every trajectory sweep point, drawn as bars and exported.	fixed
7	(misleading) Vorticity noise drawn full scale	Below 10^{-6} of j/cell drawn as zero and labelled.	fixed
8	(misleading) Example notes contradicted the app: ℓ 1.7 vs 0.58 μm , R_c “22 μm per mT” vs 117, “the sweep runs”, probe voltages “in the detail tables”, aperture not superballistic at 200 K	All corrected. Probe potentials are now listed on the Transmission tab. The aperture example runs at 300 K and says where it crosses Sharvin.	fixed
9	(misleading) Validity said “no tunnelling” while four examples are about Klein tunnelling	Validity and Assumptions rewritten: what is included, what is not.	fixed
10	(misleading) Beads in trajectory densities look like interference	Note in the Veselago and DFM examples: sampling of the stepped tracer, not interference.	documented
11	(misleading) Headline conductance switches model silently; FEM exceeds Sharvin	The headline names its model. Validity warns when the field-model G exceeds the Sharvin limit.	fixed
12	(misleading) Density colour range in m^{-2} labelled cm^{-2} ; signed n on a sequential map	Scale in cm^{-2} ; a density that changes sign uses the diverging map.	fixed
13	(misleading) $G = 0$ at $n = 0$ with no comment	Validity states that the Dirac-point physics (minimum conductivity, puddles, thermal smearing) is not modelled.	documented
14	(confusing) “Width” is the source–drain length	Labels “Length L along x ”, “Width W along y ”.	fixed
15	(confusing) Greyed ℓ field shows a stale 0.5	Shows the value in use (0.583).	fixed
16	(confusing) Negative sizes and mobility accepted	Clamped, with a message.	fixed

#	Finding	Fix / evidence	Status
17	(confusing) Legend “ballistic” for the Sharvin+Drude series	Legend, strip and CSV name the model actually used.	fixed
18	(confusing) No contact labels or p–n/obstacle outlines on the views; axis caption over the source	—	not done
19	(confusing) Colour-bar numbers only inside a collapsed group	Also shown in the status strip.	fixed
20	(confusing) Help is thin; notes use internal ids	Internal ids removed from the notes; more “?” help topics not added.	partly
21	(cosmetic) Status lags: “complete” after a failure, no “stopped”, menu “G vs” while R is plotted	Error and stop states shown; menu reads “versus ...”.	fixed
22	(cosmetic) Density field shows 1000000000000	Written as 1e12.	fixed
23	(confusing) A 3-minute EMR sweep has no time estimate	Remaining-time estimate in the sweep strip.	fixed
24	(praise) Same-kind map axes refused; $T \pm \sigma$ table; Stop works; electron optics and viscous vicinity behave as described; good references	—	—

4 Test 3: Prof. Jonnson, experimentalist

4.1 His verdict before the fixes (excerpts)

“The field (FEM) model passes. Van der Pauw gives R_s within 0.3 % of $1/(ne\mu)$ whatever the contact size or grid. [...] The Sharvin conductance is exact to four digits. [...] Results are bit-identical across thread counts and after saving and reloading a project. That is better than most homemade codes I have refereed. I would use the field model in a paper now.

I would not yet put trajectory (Monte Carlo) numbers in a paper [...]: the ‘Random seed’ does not give independent runs; deep in the diffusive regime the Monte Carlo is wrong without any warning; the multi-terminal resistances carry no error bar anywhere; and the Monte Carlo Hall resistance at $\mu B = 5$ is 14 % low on the default bar and 29 % high on a long bar with thin probes [...] a referee will ask. The exports are not ready for a methods section.”

4.2 Quantitative checks

Table 3 is a selection from his 27-row table. The “after” column was measured on the fixed core with his own harnesses, rebuilt unchanged (`jonnson/after/`), and with the new `statcheck`.

Table 3: Textbook checks before and after.

Check	Before	After
Sharvin $G = (4e^2/h)k_F W/\pi$, $n = 10^{11} - 4 \times 10^{12}$	1.0000	unchanged
Two-terminal FEM vs $\sigma W/L$ (res 200)	0.9946	unchanged
van der Pauw R_s vs $1/(ne\mu)$	0.997–1.000	unchanged

Check	Before	After
FEM Hall, default bar	0.957 of B/ne	unchanged (end-contact shorting)
Magnetic focusing, first peak	0.986 of $2\hbar k_F/eL$	unchanged
Trajectories vs Landauer–Drude, $\ell = 0.05\text{--}5\text{ }\mu\text{m}$	0.96–1.04	0.98–1.01
Trajectories vs Landauer–Drude, $\ell = 20\text{ nm}$	0.49	1.057, nothing truncated
Seeds 7 and 8: shared trajectories	1935 of 2000	0 of 2000
Seed scatter / reported binomial σ (20–24 seeds)	0.52	0.87–0.91
Error bar on a Büttiker R_{xy}	none	jackknife; seed scatter / jackknife = 1.31 (16 seeds)
Phonon ρ at $4n$ vs n , 300 K	0.50 ($\propto n^{-1/2}$)	1.00 (independent of n)
Phonon ρ below T_{BG}	linear in T	$\propto T^4$: $\ell(5\text{ K})/\ell(10\text{ K}) = 15.9$
FEM on a $4000\text{ }\mu\text{m}$ bar (from Sofia)	4× too large , “converged”	1.008 of $\sigma W/L$

4.3 The Monte Carlo Hall resistance (his finding 4)

His two numbers, 0.86 and 1.29 of B/ne , came from runs of 4000 trajectories, where the seed-to-seed spread is 8.9%, and with seeds that were not independent. Re-measured with 80 000 trajectories, jackknife errors and decreasing mean free path (default bar, $B = 0.5\text{ T}$, $r_c = 0.233\text{ }\mu\text{m}$):

ℓ (μm)	MC R_{xy} (Ω)	FEM (Ω)	B/ne (Ω)	MC/FEM
1.000	316.7 ± 3.8	303.8	312.1	1.042
0.583	314.2 ± 5.1	299.4	312.1	1.049
0.300	295.4 ± 6.8	292.9	312.1	1.009
0.150	284.0 ± 19.6	287.4	312.1	0.988
0.080	194.6 ± 50.4	285.0	312.1	0.68 (1.8 σ)

In the diffusive limit the Monte Carlo goes to the FEM value within its error bars. In the quasi-ballistic regime it sits about 4% above, at B/ne , because ballistic carriers short the Hall voltage at the end contacts less. That is physical, not a fault. Near $\ell \rightarrow 0$ the multi-terminal estimate becomes very noisy, as he noted; the jackknife error now says so.

4.4 Findings and what was done

Table 4: Test 3 findings.

#	Finding	Fix / evidence	Status
1	(bug, high) Seeds not independent	The seed is hashed before it meets the trajectory index. 0 shared trajectories; the old mixing fails the same gate.	fixed
2	(bug, high) $240\text{ }\mu\text{m}$ path cap biased the diffusive T low by 30–50%; $G = 0$ at $\ell \leq 10\text{ nm}$, silently	Path budget grows to $20L^2/\ell_{tr}$ (a diffusive carrier needs about L^2/ℓ). Truncated carriers are counted and warned about in the run, the sweep and the CSV.	fixed

#	Finding	Fix / evidence	Status
3	(missing, high) No error bars on multi-terminal resistances, sweeps or CSV	Every trajectory belongs to one of 10 batches; leave-one-out jackknife on R_{xy} , R_{xx} and G . Matches the seed scatter (statcheck).	fixed
4	(possible wrong physics) MC Hall -14% / $+29\%$	Statistics and correlated seeds; converges to the FEM value in the diffusive limit (table above).	not a bug
5	(misleading) G shown to one significant figure; model not named	Three significant figures; the headline names its model and error.	fixed
6	(export, high) CSV had no units, model, N , seed, σ or version; JSON had no units	CSV header block: measure, model, trajectories per point, seed, quality, device, units in every column name, error and truncation columns. JSON carries units. A build identifier is not yet exported.	partly
7	(misleading) “ballistic” label on diffusive results; “Sharvin + Drude” on a Hall sweep	Labels name the model actually used.	fixed
8	(documentation) Notes: ℓ $1.7\mu\text{m}$; focusing between facing probes; “full-width” contacts	Corrected ($0.58\mu\text{m}$; focusing needs two contacts on one edge; 90% of the edge).	fixed
9	(wrong physics) Phonon $\rho \propto n^{-1/2}$, no Bloch–Grüneisen regime	$\ell_{ph} \propto 1/k_F$ (Hwang & Das Sarma 2008), $\rho \propto T^4/(T^3 + T_{BG}^3)$ with $T_{BG} = 2\hbar v_s k_F / k_B$ (Efetov & Kim 2010). Off by default, so nothing old moves.	fixed
10	(missing physics, undocumented) Nothing at the Dirac point	Stated in Validity and Assumptions.	documented
11	(documentation) e–e constant $C = 1$ uncalibrated, no log factor	Stated in Validity; calibrate before quoting a viscosity.	documented
12	(documentation) Ideal contacts at cell centres; effective length	Contact model stated in Validity.	documented
13	(missing) One edge condition for all edges	—	not done
14	(documentation) References: one unverifiable, classics missing	Unverifiable arXiv entry dropped; Beenakker & van Houten (1991) and Taychatanapat et al. (2013) added.	fixed
15	(praise) Sharvin, vdP, R_{xx} , FEM Hall, focusing, reproducibility, honest $T \pm \sigma$	—	—

5 What changed in the program

- Core.**
- Hashed seeds.
 - Diffusive path budget.

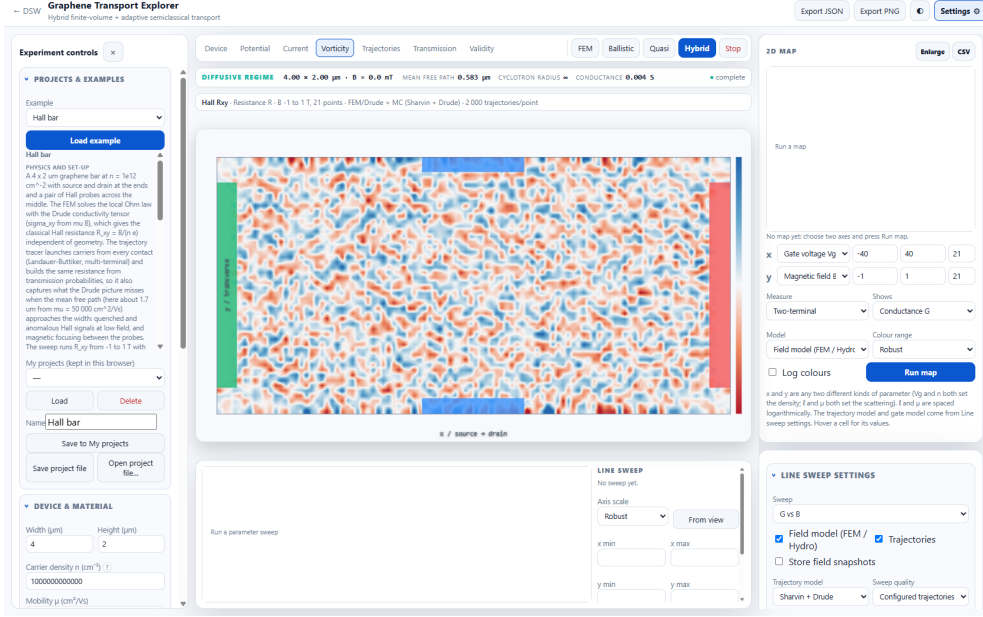


Figure 1: Before the fix: the Vorticity tab of an Ohmic Hall bar at $B = 0$. The pattern is solver round-off (maximum $|\omega|$ is 2.4×10^{-16} of the current density per cell). It is now drawn as zero, and the colour bar reads “vorticity (none above round-off)”. At $B \neq 0$ the vorticity at the contact corners is real (ratio 0.55) and is still drawn.

- Ten jackknife batches per run, with errors on every sweep point.
- Strict FEM convergence with the direct fallback; default tolerance 10^{-9} .
- Phonon law with $\ell_{ph} \propto 1/k_F$ and a Bloch–Grüneisen regime.
- No zero probe rows in single runs.
- Density scale in cm^{-2} , with a diverging map for signed densities.
- Round-off vorticity drawn as zero.
- Test hooks (`seedMix:0`, `pathAuto:0`), so the identity gate can still compare against the old engine.

- Panel.**
- Error bars and model-named legends.
 - Truncation warnings and a remaining-time estimate.
 - Self-describing CSV and JSON.
 - Stale-results marker; loading a project stops and clears.
 - Headline conductance with its model and error.
 - Validity and Assumptions rewritten; Sharvin-limit warning.
 - Probe potentials on the Transmission tab.
 - Dimension labels, input checks, number formatting, error and stop states.

Examples. Every claim in the notes checked against the program; internal ids replaced by panel names; the aperture runs where it is superballistic; references corrected.

New gates. `test/statcheck.cpp` covers seed independence, honest binomial and jackknife errors, diffusive Landauer–Drude, and FEM convergence on a long bar. Each check is shown to fail on the old behaviour. `DSW/tools/probe-transim-tests.js` runs 20 live checks of the panel fixes.

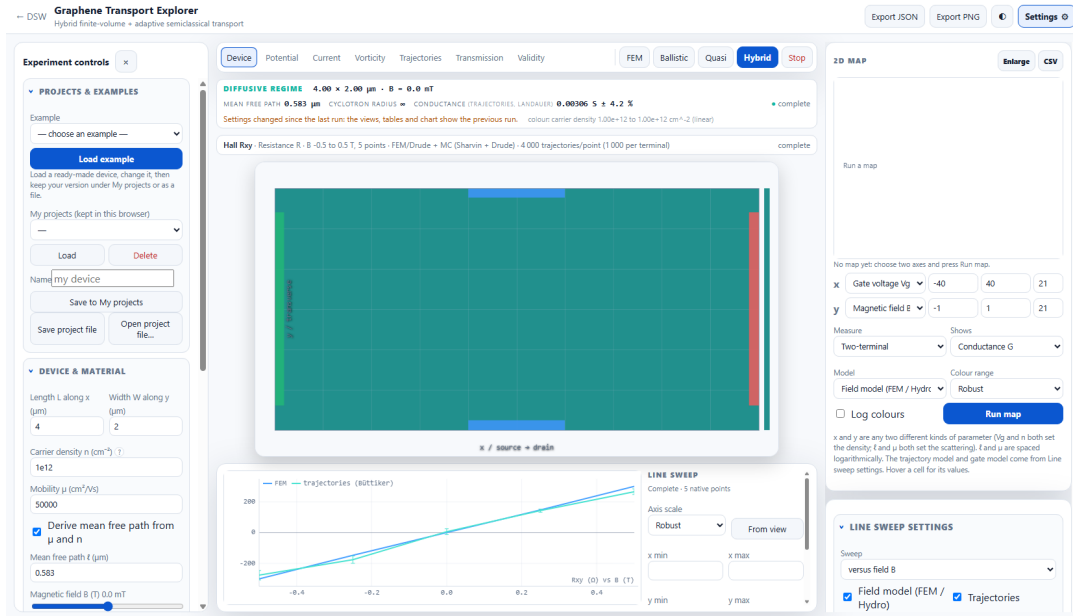


Figure 2: After the fixes: a Hall sweep with the field model and the Büttiker trajectories. Every trajectory point carries its jackknife error bar. The headline conductance names its model and its error, the status strip shows the colour range, and the stale-results note appears because a setting was changed after the last single run.

6 What is still open

- Contact labels and p–n/obstacle outlines on the views (Sofia 18), and more “?” help topics (Sofia 20).
- A different boundary condition per edge (Jonsson 13), needed to see the second and third focusing peaks.
- Dirac-point physics (minimum conductivity, puddles, thermal smearing) and a calibrated electron–electron length. Both are now stated as absent, not implemented.
- A build identifier in the exports.

7 Verdict after the fixes

The field model was already fit for publication-grade geometry corrections (Hall bar, van der Pauw, EMR, contact shorting) and remains so. The trajectory model’s three blocking faults are fixed: seed independence, the diffusive bias, and missing error bars. Its error bars are now shown to be honest against independent seeds, and its exports describe themselves. Its numbers can go into a methods section with the stated caveats: semiclassical and phase-incoherent, ideal contacts, nothing at the Dirac point. For a learner, the program now says which model a number comes from, how noisy it is, and when a result belongs to settings that have since changed. Those were the three things that misled the student.